

Midterm Examination

Due: Thursday, October 15 at 10:00 AM

Value: 180 points

Format: 75-minute in-class examination

Purpose

Demonstrate conceptual understanding of research methods, perception, attention, working memory, and long-term memory by applying course ideas to unfamiliar evidence.

Your task

- Section A (40): Answer 8 of these 10 terms in 2-3 sentences and give an original example: signal detection, attentional capture, conjunction search, change blindness, phonological loop, central executive, encoding specificity, retrieval cue, proactive interference, and consolidation. Each response is worth 5 points.
- Section B1 (10): From the supplied table, identify the independent variable and both dependent measures.
- Section B2 (10): Calculate the dual-task accuracy cost for low and high memory load.
- Section B3 (10): Describe the joint pattern in accuracy and response time without explaining it.
- Section B4 (10): Use a working-memory account to explain why the dual-task cost differs by load.
- Section B5 (10): Evaluate the claim that high memory load directly causes every observed difference; name one uncontrolled alternative.
- Section C (60): Answer 3 of 4 cases, 20 points each: (1) a confident eyewitness identifies a briefly seen stranger after a suggestive lineup; (2) a phone redesign groups alerts but adds a persistent badge; (3) a student rereads fluent notes but performs poorly on a delayed test; (4) a nurse must retain a dosage while responding to an interruption. For each, name two relevant concepts, explain the mechanism, and recommend one evidence-based change.
- Section D (30): Design a controlled experiment testing whether reinstating a study context improves delayed recall. Specify participants, materials, independent and dependent variables, assignment, controls, predicted result, one confound, and one ethical consideration.

Provided evidence or data

Condition	Single-task accuracy	Dual-task accuracy	Mean RT
Low memory load	94%	88%	610 ms
High memory load	91%	69%	845 ms

Submit

- Handwritten responses in the examination booklet.
- One annotated instructor-provided concept sheet, submitted with the exam.

Important note

Closed note except for the approved concept sheet. Calculators are not required. Students with testing accommodations should confirm arrangements by October 8.

Grading criteria

Criterion	Pts	Full-credit evidence
Section A	40	Accurate definitions paired with original, diagnostic examples.
Section B	50	Correctly reads the graph and separates description from inference.
Section C	60	Applies mechanisms to three cases with explicit reasoning.
Section D	30	Specifies variables, controls, predicted result, and limitation.